

Sabina Sagynbayeva

Curriculum Vitae



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RESEARCH INTERESTS

I am an astrophysicist who primarily works on the past, present, and future of exoplanets – I study planetary dynamics and planet formation to understand the architecture of exoplanetary systems. In parallel, I also work on stars that host those exoplanets and stellar characterization by studying their surface activity using the time-series data from *Kepler* and *TESS* telescopes. I am broadly interested in general analytical techniques, astro-statistics, and numerical simulations.

EDUCATION

- 2021 - 2026 **PhD:** Physics (Astrophysics)
Stony Brook University, Stony Brook, NY
- 2021 - 2023 **MA:** Physics (Astrophysics)
Stony Brook University, Stony Brook, NY
- 2016 – 2020 **Bachelor of Science:** Physics
Minor: Mathematics
Nazarbayev University, Astana, Kazakhstan

RESEARCH POSITIONS

- 2026 - PRESENT **McGill University:** CITA National Fellow
- 2026 - PRESENT **McGill University:** Trottier Space Institute Fellow
- 2021 - 2026 **Stony Brook University:** Graduate Student Researcher
Advisors: Phil Armitage, Will Farr
- 2022 - 2026 **Flatiron Institute:** Guest Researcher
- 2025 **Kavli Institute for Theoretical Physics:** Graduate Research Fellow
Advisor: Lars Bildsten
- 2019 **University of Cambridge, UK:** Research Intern
Advisor: Roman Rafikov
- 2016-2021 **Nazarbayev University:** Research Assistant
Advisor: Daniele Malafarina

AWARDS, GRANTS, & FELLOWSHIPS

- Raynor L. Duncombe Student Research Prize** 2026
The prize awarded annually by the Division on Dynamical Astronomy.
- KITP Graduate Fellowship** 2024-2025
I was selected for the opportunity for advanced physics doctoral students to spend a minimum period of 5 months at the Kavli Institute for Theoretical Physics.
- Peter Kahn Prize** 2024
An award for "outstanding research" for which I was nominated by my academic advisor.
- The Other Worlds Laboratory Exoplanet Summer Program** 2023
I was selected for the program that allows to visit UC Santa Cruz for three weeks to work on a project with an UCSC faculty.
- Frontera Computational Science Fellowship** 2023 – 2024
1-year fellowship for graduate students with an opportunity to compute on Frontera.
- LSSTC Data Science Fellowship Program** 2022 – 2024
I was selected for the program that consists of six week-long sessions on data science.
- Young Researchers Alliance FRIP program** 2020
The stipend awarded to students for research projects. Stipend: \$1,000
- Yessenov Foundation Scholarship** 2019
Awarded to ten best students from Kazakhstan for a research internship in the US and European universities and laboratories.
Funding: \$7,500

SELECTED INVITED (16) & CONFERENCE TALKS

Exoplanets VI, Porto, Portugal	Jul 2026
57th DDA meeting, Chicago, Illinois	Jun 2026
Physics Colloquium , Queens College	Apr 2026
IREx Cafe, University of Montreal	Jan 2026
CIERA Observer's seminar, Northwestern University	Dec 2025
Exoplanets, Stars, and Planet Formation, Space Telescope Science Institute	Dec 2025
Carnegie EPL Astronomy Seminar, Carnegie Institution for Science	Oct 2025
Stars & Planets workshop, MIT	Aug 2025
Exoplanets group meeting, Princeton University	Jun 2025
Exoplanets group meeting, Queen Mary University, London, UK	Jun 2025
Dynamix Conference, Cambridge, UK	Jun 2025
56th DDA meeting, Atlanta, Georgia	May 2025
Planet Formation and Migration near the Inner Edge of Disks, KITP Program	Apr 2025
A&A Journal Club Talk, University of California San Diego	Apr 2025
AstroLunch seminar talk, University of California Santa Barbara	Mar 2025
PLUNCH seminar talk, University of California Santa Cruz	Mar 2025
KITP Local's Lunch, KITP	Feb 2025
Planet Formation group meeting, Flatiron Institute (CCA)	Feb 2025
MAPL Lab Group Meeting, University of California Santa Barbara	Feb 2025
Planet Formation group meeting, Flatiron Institute (CCA)	Oct 2024
Frontera Talk, Texas Advanced Computing Center	May 2024
New York Area Exoplanets Meeting (NYAEM) 2024	May 2024
Lunch Talk, Columbia University	Feb 2024
Bay Area Exoplanet Meeting 44, University of California Santa Cruz	Jul 2023
University of California Santa Barbara	Jul 2023
OWL talk, University of California Santa Cruz	Jul 2023
StanCon 2023, Washington University in St. Louis	Jun 2023
Athena++ workshop, Flatiron Institute (CCA)	May 2023
Gravitational Waves group meeting, Flatiron Institute (CCA)	Oct 2022
Seminar, University of Kansas	May 2022

TEACHING APPOINTMENTS

	MAR 2026
Guest Lecturer	
Course: PHY 390 Special Topics in Astrophysics	MAY 2023 – JUL 2023
Teaching Assistant	
Course: Classical Physics Lab	
Department of Physics & Astronomy, Stony Brook University	MAR 2022 – APR 2022
Group Project Leader	
Women in Science and Engineering program	
Stony Brook University	AUG 2021 – DEC 2021
Teaching Assistant	
Course: Introduction to Planetary Sciences	
Department of Physics & Astronomy, Stony Brook University	JAN 2017 – JAN 2019
Teaching Assistant	
Courses: Calculus I,II,III, Linear Algebra, Ordinary Differential Equations	
Department of Mathematics, Nazarbayev University	

ACADEMIC LEADERSHIP AND SERVICE

Reviewer PASA, JOSS	MAY 2024 – MAY 2025
Steering Committee Member NASA HWO Demographics and Architectures Sub-WG	OCT 2023
Executive Secretary NASA Astrophysics Theory Program (ATP)	SEP 2021 – MAY 2023
Senator for the Department of Physics & Astronomy, Member of the Graduate DEI Committee Graduate Student Organization, Stony Brook University	JUN 2021 – JUN 2022
Underclass person-at-large & Director of External Affairs Physics Graduate Student Association, Department of Physics & Astronomy, Stony Brook University	SEP 2017 – MAY 2020
Physics Department Representative Student Council of Nazarbayev University	

SELECTED OUTREACH : WATCHABLE TALK

Outreach talk: The formation of gas giants 	MAR 2023
iTelescope Webinar Series	AUG 2022
Outreach talk: Oceans in the Solar System	APR 2021
Astronomy on Tap, New York City	
Outreach talk: How do planets form? 	JAN 2021 – JAN 2023
Astronomy on Tap, Baton Rouge	
Writer for Astrobites.org	APR 2018 – MAY 2020
A website where graduate students publish daily summaries of recent papers on astro-ph. I also chaired the Advertising and Undergraduate Committees.	
President of the Women in Physics Club	OCT 2017 – SEP 2020
Nazarbayev University	
Organizer at the “Education for all” center	
An organization that helps children with mental and physical disabilities. I organized the first three inclusive musical theatre performances in Kazakhstan	

COMPUTATIONAL SKILLS

PROGRAMMING / MARKUP LANGUAGES	Python, Julia, C/C++, IDL, HTML, JavaScript, Mathematica, $\LaTeX$
STATISTICAL SKILLS	Hierarchical Bayesian Models, Gaussian Processes, MCMC sampling
HYDRO/N-BODY CODES	Athena++, PLUTO, FARGO3D, REBOUND
FRAMEWORKS / TOOLS	git, GitHub, ds9, Slurm
SUPERCOMPUTING CLUSTERS	<i>seawulf</i> at SBU, <i>Frontera</i> at the Texas Advanced Computing Center, <i>rusty</i> at Flatiron Institute

PUBLICATIONS: 1ST/2ND AUTHOR : IN PRESS (CLICKABLE)

15. **Sabina Sagynbayeva** et al. “Rotation Periods for Stars in Open Cluster NGC 6819 From Kepler IRIS Light Curves.” *In review; on arXiv*.

14. **Sabina Sagynbayeva** et al. “Polka-dotted Stars II: Starsposts and Obliquities of Kepler-17 and Kepler-63.” *The Astrophysical Journal* (2026).

13. **Sabina Sagynbayeva** et al. “Polka-dotted Stars: a Hierarchical Model for Mapping Stellar Surfaces Using Occultation Light Curves.” *The Astrophysical Journal* (2025). 

12. **Sabina Sagynbayeva** et al. “Requirements for Joint Orbital Characterization of Cold Giants and Habitable Worlds with Habitable Worlds Observatory.” *The Astronomical Journal* (2025).

11. **Sabina Sagynbayeva** et al. “Circumplanetary Disks are Rare around Planets at Large Orbital Radii: A Parameter Survey of Flow Morphology around Giant Planets.” *The Astrophysical Journal* (2025).

10. Daniele Malafarina, **Sabina Sagynbayeva** (equal contribution). “What a difference a quadrupole makes?” *General Relativity and Gravitation* (2021).

9. Courtney Dressing et al. [including **S. Sagynbayeva**]. “Early Exploration of the Scientific Discovery Space for the Habitable Worlds Observatory.” *On arxiv* (2026).
8. Stephen Kane et al. [including **S. Sagynbayeva**]. “Imaging Venus-like Worlds: Spectral, Polarimetric, and UV Diagnostics for the Habitable Worlds Observatory” *Publications of the Astronomical Society of the Pacific* (2026).
7. Mona El Morsy et al. [including **S. Sagynbayeva**]. “OASIS Survey Direct Imaging and Astrometric Discovery of HIP 71618 B: A Substellar Companion Suitable for the Roman Coronagraph Instrument Technology Demonstration” *The Astrophysical Journal Letters* (2025).★
6. Thayne Currie et al. [including **S. Sagynbayeva**]. “SCEXAO/CHARIS and Gaia Direct Imaging and Astrometric Discovery of a Superjovian Planet 3-4 λ/D from the Accelerating Star HIP 54515” *The Astronomical Journal* (2025). ★
5. Sarah Blunt et al. [including **S. Sagynbayeva**]. “A Statistical Method for Constraining the Capability of the Habitable Worlds Observatory to Understand Ozone Onset Time in Earth Analogs.” *JATIS* (2025).
4. Rachel B. Fernandes et al. [including **S. Sagynbayeva**]. “Signatures of Atmospheric Mass Loss and Planet Migration in the Time Evolution of Short-Period Transiting Exoplanets.” *The Astronomical Journal* (2025). ★
3. Briley Lewis et al. [including **S. Sagynbayeva**]. “Exploring the Effects of Astrobites Lesson Plans on Undergraduate Astronomy Students.” *Physical Review Physics Education Research* (2025).
2. Thayne Currie et al. [including **S. Sagynbayeva**]. “Direct Imaging and Astrometric Discovery of a Superjovian Planet Orbiting an Accelerating Star.” *Science* (2023). ★
1. D. Alina et al. [including **S. Sagynbayeva**]. “Large-scale magnetic field in the Monoceros OB-1 East molecular cloud.” *Astronomy & Astrophysics* (2020).